QUOC ANH TRAN

Department of Mathematics ◇ University of Chicago
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EDUCATION

Rice University, Houston, TX 2021
B.S. in Mathematics
University of Illinois Chicago, Chicago, IL 2022 - 2026
University of Chicago, Chicago, IL 2026 - (expected) 2027
Ph.D. in Mathematics

RESEARCH INTERESTS

Algebraic geometry, hodge theory, non-abelian hodge theory, twistor/hodge modules, deformation theory, hodge theoretic moduli and positivity.

PAPERS

On the nilpotent residue non-abelian Hodge correspondence for higher-dimensional quasiprojective varieties
submitted, 2026, https://arxiv.org/abs/2601.15553

Perverse Euler Characteristics of Hermitian Locally Symmetric Spaces
with Mert Akdenizli, submitted, 2026, https://arxiv.org/abs/2609.10946

The nonabelian Hodge correspondence for extensions on quasiprojective curves
preprint, 2026, https://arxiv.org/abs/2609.26896

Rank two cohomology jump loci on projective varieties
preprint, 2026, https://arxiv.org/abs/2610.36251

ORGANIZATIONAL EXPERIENCE

Organizer, Satake-Baily-Borel and toroidal compactifications reading group Spring 2026
Organizer, Kobayashi-Hitchin correspondence reading group Spring 2025
Organizer, Moduli of K3 surfaces reading group Fall 2024
Co-organizer, Graduate Algebraic Geometry seminar 2024-2025
Co-organizer, Graduate Number Theory seminar Spring 2024

HONORS, AWARDS

Hubert E. Bray Prize, given annually to the most outstanding mathematics major at Rice.

GRADUATE SEMINAR TALKS

Graduate AG Seminar. Moduli of K3 surfaces – General curve of genus 6 – Decompositions in derived categories – Generic Torelli theorem – Formal GAGA – Connectedness of Brill-Noether loci – Du Bois singularities – Stability of isomonodromic deformations – Fourier transform on elliptic curve.
Graduate NT Seminar. Root systems of algebraic groups – Jacobian varieties.
Hodge Theory Seminar. Mumford-Tate group – Deligne's finiteness theorem – Structure of period map

– Limit Mixed Hodge Structure – Kummer surfaces – Basics of D-modules – Intersection cohomology – Hodge modules over a curve – PEL and Hodge type Shimura varieties – Hodge-generic compact subvarieties of A_g – Intro to p-curvature conjecture.

CONFERENCES, WORKSHOPS

Moduli of K3 surfaces – ICERM	Nov 2026
Moduli of Varieties – SLMath Summer School	July 2026
Research seminar – Special Year at Institute for Advanced Study	November 2025
Summer Research Institute in Algebraic Geometry – Colorado State University	July 2025
Hodge theory and o-minimality – CIRM, Luminy	January 2024
Unlikely Intersection – Arizona Winter School	March 2023

TEACHING EXPERIENCE

University of Chicago

- Graduate Student Lecturer for MATH 15200 Calculus 2 — Autumn 2026

University of Illinois Chicago

- TA for MATH 160 Finite Mathematics for Business — Fall 2025, Spring 2026
- TA for MATH 125 Elementary Linear Algebra — Fall 2022 - Summer 2024, Spring 2025
- TA for MATH 210 Multivariable Calculus — Fall 2024

Rice University

- TA for MATH 354 Honors Linear Algebra, 356 Group Theory, 376 Algebraic Geometry, STAT 312 Statistics and COMP 382 Algorithms.

MENTORSHIP

University of Illinois Chicago

- Directed Reading Program — Spring 2026

  Guided Ted Nguyen through Kirillov's notes on Lie groups and algebras.